

Sanjeev Kumar

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LinkedIn | GitHub | LeetCode |

EDUCATION

IILM University, Greater Noida

B.Tech in Computer Science and Engineering

2023 - 2027

CGPA: 7.89/10

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript

Machine Learning & AI: Supervised Learning, Feature Engineering, Data Preprocessing, Model Evaluation, Explainable AI (SHAP), Scikit-learn, NumPy, Pandas, Prompt Engineering

Frameworks Technologies: Django, Next.js, React, Node.js, Express.js, MongoDB (MERN Stack), REST APIs, Tailwind CSS

Databases: MongoDB, PostgreSQL, SQL

Tools & Platforms: Git, GitHub, VS Code, Postman, Google Colab, Overleaf

Core CS & Security: Data Structures Algorithms, Database Management Systems, Operating Systems, Computer Networks

PROJECTS

INORA - Multi-Channel EEG-Based Control

Dec 2025 - Jan 2026

Technologies: AD620, TL072, ADS1115, ESP32, Python, NumPy

- Built a low-cost 3-channel EEG acquisition system with wireless streaming at **83 Hz/channel**, and developed a Python pipeline for EEG acquisition, filtering, and DRL-based noise reduction, validated on **20 volunteers**.
- Demonstrated EEG-based Pong control with valid **15–80 uV** signals using hardware costing **INR 3,146**.

BrainScore - EEG-Based Depression Detection

Mar 2025 - Nov 2025

Technologies: Python, Pandas, Scikit-learn, LightGBM, SHAP, Google Colab

- Processed EEG recordings from **58 subjects** using **20-channel** signals and extracted **18** statistical, frequency, and entropy-based biomarkers; built and evaluated **6 machine learning models**.
- Achieved **95.28%** classification accuracy using **LightGBM**, with **SHAP** analysis to interpret feature importance and improve model explainability for automated depression detection.

MUNC Dashboard - Kasper Infotect User Dashboard

June 2025 - Aug 2025

Technologies: JavaScript, Next.js, React.js

- Developed and enhanced responsive dashboard pages using **JavaScript**, **Next.js**, and **React.js**, implementing frontend features and improving UI usability across multiple pages.
- Resolved application bugs and supported testing and documentation to improve reliability and maintainability.

EXPERIENCE

Frontend React Developer - Intern

Dec 2025 - Feb 2026

Kasper Infotech Pvt. Ltd.

Remote

- Developed and improved responsive web pages using **React.js**, built reusable UI components, and enhanced interface structure, usability, and responsiveness.
- Resolved UI and frontend bugs while maintaining technical documentation and project updates.

Machine Learning & Data Science Intern

May 2025 - Jul 2025

Svaarogyam Medical Devices Pvt. Ltd.

Remote

- Worked on healthcare machine learning tasks involving **data cleaning**, **preprocessing**, **feature extraction**, and predictive modeling for medical device applications.
- Built end-to-end **ML pipeline** collaborated with a remote team to deliver project milestones in medical domain.

ACHIEVEMENTS / CERTIFICATIONS

- Published Patent Application, Indian Patent Office (Application No. 202611071187): Wearable Head-Mounted Device for Continuous Mental State Monitoring and Adaptive Cognitive Regulation
- Research article accepted in IEEE Transactions on Medical Robotics and Bionics (T-MRB): SHAP-based Study: XAI-Guided Optimisation of Spatial-Temporal Bio-Neuro Marker EEG for Depression
- Research Paper accepted at IEEE Sponsored ETNCC 2026 for INORA: A Low-Cost 3-Channel EEG Acquisition System with Driven-Right-Leg Noise Suppression for BCI Prototyping
- Hackathon Winner (1st Rank) - Secured 1st place among 1,004 participants (118 teams)
- Secured 1st place in the Call for Paper event at SRM Institute of Science and Technology